

Day – 9 Inter – VLANs Routing Basics Theory.

Task – 1 Create three VLANs

- configure terminal
 - interface FastEthernet0/2
 - switchport mode access
 - switchport access vlan 10
 - interface FastEthernet0/3
 - switchport mode access
 - switchport access vlan 20
 - interface FastEthernet0/4
- Here are the step-by-step CLI commands to create your application-themed VLANs (**Foodies**, **Shoppers**, and **Streamers**) on your Cisco switch and assign PCs to each respective segment.
- - **### Step 1: Create and Name the VLANs on the Switch**
 - Open your switch CLI, go to global configuration mode, and create the three VLANs:
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 - `enable`
 - `configure terminal`
 - `vlan 10`
 - `name Foodies`
 - `vlan 20`
 - `name Shoppers`
 - `vlan 30`
 - `name Streamers`
 - `Exit`
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 - **Step 2: Assign Switch Ports to the VLANs**
 - Assume you connect your three PCs to ports FastEthernet0/1, FastEthernet0/2, and FastEthernet0/3. Assign them as access ports to their respective VLANs:
 - Plaintext
 - `interface FastEthernet0/1`
 - `switchport mode access`
 - `switchport access vlan 10`
 - `interface FastEthernet0/2`

- switchport mode access
- switchport access vlan 20
- interface FastEthernet0/3
- switchport mode access
- switchport access vlan 30
- End
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- Step 3: Configure the PCs
- Connect a PC to each of those ports in Packet Tracer and assign them matching IP addresses for their respective subnets:
- PC-Foodies (Zomato simulation): Connected to Fa0/1
- IP Address: 192.168.10.10 | Subnet Mask: 255.255.255.0
- PC-Shoppers (Flipkart simulation): Connected to Fa0/2
- IP Address: 192.168.20.10 | Subnet Mask: 255.255.255.0
- PC-Streamers (Spotify simulation): Connected to Fa0/3
- IP Address: 192.168.30.10 | Subnet Mask: 255.255.255.0

Task – 2 Config the router in the topology.

- Step 1: Switch Configuration
- First, create the VLANs, assign the end-device access ports, and configure the uplink trunk port connected to the router.
- Create VLANs 10 (Foodies), 20 (Shoppers), and 30 (Streamers).
- Assign switch ports to their respective VLAN access modes.
- Configure the switch port connected to the router as an 802.1Q trunk port.
- Switch> enable
- Switch# configure terminal
- Switch(config)# vlan 10
- Switch(config-vlan)# name Foodies
- Switch(config-vlan)# vlan 20
- Switch(config-vlan)# name Shoppers
- Switch(config-vlan)# vlan 30
- Switch(config-vlan)# name Streamers
- Switch(config-vlan)# exit
- ! Assign access ports (adjust port numbers as needed for your topology)
- Switch(config)# interface FastEthernet0/1
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 10
- Switch(config)# interface FastEthernet0/2

- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 20
- Switch(config)# interface FastEthernet0/3
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 30
- Configure Trunk Port to Router
- Switch(config)# interface FastEthernet0/24
- Switch(config-if)# switchport mode trunk
- Switch(config-if)# end
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- Step 2: Router Configuration
- Next, enable the physical interface on the router and configure the sub-interfaces with IEEE 802.1Q encapsulation and gateway IP addresses.
- Enable the physical interface (GigabitEthernet0/0).
- Create sub-interfaces .10, .20, and .30 matching the VLAN IDs.
- Assign encapsulation tags and default gateway IP addresses.
- Router> enable
- Router# configure terminal
- Router(config)# interface GigabitEthernet0/0
- Router(config-if)# no shutdown
- Router(config-if)# interface GigabitEthernet0/0.10
- Router(config-subif)# encapsulation dot1q 10
- Router(config-subif)# ip address 192.168.10.1 255.255.255.0
- Router(config-subif)# interface GigabitEthernet0/0.20
- Router(config-subif)# encapsulation dot1q 20
- Router(config-subif)# ip address 192.168.20.1 255.255.255.0
- Router(config-subif)# interface GigabitEthernet0/0.30
- Router(config-subif)# encapsulation dot1q 30
- Router(config-subif)# ip address 192.168.30.1 255.255.255.0
- Router(config-subif)# end
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- Step 3: PC IP Configuration & Verification
- Configure each PC's desktop IP settings in Packet Tracer with the corresponding subnet and default gateway:
- Foodies PC (VLAN 10): IP: 192.168.10.10 | Subnet Mask: 255.255.255.0 | Default Gateway: 192.168.10.1
- Shoppers PC (VLAN 20): IP: 192.168.20.10 | Subnet Mask: 255.255.255.0 | Default Gateway: 192.168.20.1
- Streamers PC (VLAN 30): IP: 192.168.30.10 | Subnet Mask: 255.255.255.0 | Default Gateway: 192.168.30.1

Task – 3 Setting the gateway to pc.

- Quick Assignment Checklist
- Foodies PC (VLAN 10)
- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- Shoppers PC (VLAN 20)
- IP Address: 192.168.20.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.20.1
- Streamers PC (VLAN 30)
- IP Address: 192.168.30.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.30.1

Task – 4 Solving the scenario of router.

- Step 1: Diagnose and Fix the Router Sub-Interface Configuration
- A frequent mistake is missing the encapsulation dot1q [vlan-id] command or applying the wrong VLAN ID to a sub-interface, which causes the router to drop or misroute tagged frames.
- How to check: Run show ip interface brief and show running-config on the router. Ensure that the sub-interfaces match the correct VLAN numbers and have the right encapsulation tags.
- Router> enable
- Router# configure terminal
- Router(config)# interface GigabitEthernet0/0.20
- Router(config-subif)# encapsulation dot1q 20
- Router(config-subif)# ip address 192.168.20.1 255.255.255.0
- Router(config-subif)# interface GigabitEthernet0/0.30
- Router(config-subif)# encapsulation dot1q 30
- Router(config-subif)# ip address 192.168.30.1 255.255.255.0
- Router(config-subif)# end
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- Step 2: Verify and Fix Switch Trunk Port Settings
- If the switch port connected to the router (FastEthernet0/24) is left as an access port instead of a trunk port, tagged frames from multiple VLANs will never reach the router's sub-interfaces.

- How to check: Run show interfaces trunk on the switch. If no trunk ports are listed, that is your culprit.
- Switch> enable
- Switch# configure terminal
- Switch(config)# interface FastEthernet0/24
- Switch(config-if)# switchport mode trunk
- Switch(config-if)# end
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- Step 3: Check Access Port VLAN Assignments
- Ensure that the PC representing Flipkart (Shoppers) is plugged into a switch port assigned to VLAN 20, and the PC representing Spotify (Streamers) is plugged into a port assigned to VLAN 30.
- How to check: Run show vlan brief on the switch to verify port-to-VLAN mappings.
- Switch> enable
- Switch# configure terminal
- ! Fix Shoppers PC port (e.g., Fa0/2) if it's on the wrong VLAN
- Switch(config)# interface FastEthernet0/2
- Switch(config-if)# switchport access vlan 20
- ! Fix Streamers PC port (e.g., Fa0/3) if it's on the wrong VLAN
- Switch(config)# interface FastEthernet0/3
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 30
- Switch(config-if)# end

Task - 5 Layer 3 switch and Router pros and cons.

- Approach	- Advantages	- Disadvantages
- Router-on-a-Stick (Roas)	- 1. Cost-Effective: Requires only a standard Layer 2 switch and a single physical router interface. - 2. Simple Setup: Straightforward configuration ideal for smaller networks with limited VLANs.	- 1. Bandwidth Bottleneck: All inter-VLAN traffic shares a single physical link, limiting throughput. - 2. Higher Latency & CPU Load: Traffic must traverse the trunk link twice per packet, and

- Approach	- Advantages	- Disadvantages
		the router handles all processing.
- Layer 3 Switch (SVI)	- 1. High Performance: Routing occurs at wire speed using hardware-based ASICs. - 2. No Single Bottleneck: Eliminates the physical uplink constraint since routing happens internally across the switch backplane.	- 1. Higher Cost: Layer 3 switches are significantly more expensive than standard Layer 2 switches. - 2. Increased Complexity: Configuration and troubleshooting involve managing multiple Switch Virtual Interfaces (SVIs) and routing features.

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